CarboNature

VERIFIED CREDITS FACTORY

AI Soil MRV Module & Agent Department

The build roadmap for CarboNature's Verra **VM0042 v2.2** carbon-credit engine — from field sampling and soil-carbon modelling to an autonomous AI-agent department that drives credits to issuance in the shortest possible time.

TIER 1 · THIS BUILD

AI-Driven MRV Module

Sampling, lab ingestion, DNDC/DayCent modelling, GHG accounting and VM0042 compliance — on the live database.

TIER 2 · NEXT

Verified Credits Factory

Five AI agents (John, Reveka, Dave, Ron, Jennifer) with a control-tower dashboard and long-term memory.

TIER 3 · LATER

Hardening & scale

Non-functional targets, in-house CRM, unified admin, multi-project scale.

Document Functional Specification v2.0 **Supersedes** AI Soil Module Spec v1.0 (Apr 2026)

Owner Nitzan Bauer **Audience** Engineering / AI-tooling team **Date** 28 July 2026

Methodology VM0042 v2.2 · ICVCM CCP

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0 Document control & what changed in v2.0

This revision rebuilds the April 2026 lean specification against two things that did not exist then: a **complete, live database** (41 tables on AWS RDS, seven build stages verified) and a **decision to build an autonomous AI-agent department** on top of the MRV module. Every screen, field list, and compliance hook here is reconciled with the database that is already running.

Area	v1.0 (April 2026)	v2.0 (this document)
Structure	One "lean" module, 9 screens, 3 priority tiers	Two product tiers — **Tier 1 MRV module** and **Tier 2 Verified Credits Factory** — plus a Tier 3 hardening backlog
Quantification	QA1 & QA2	**QA1 + QA2 + QA3**, full VM0042 v2.2 alignment, ICVCM CCP label
Composites / stratum	≥ 3	**≥ 5** composites per stratum changed
MCP sampler token	window + 7 days	window + **14 days**, configurable changed
Forecast dashboard (old §6.8)	"Forecast vs Actuals"	Replaced by the **Agents Dashboard** (Tier 2, Part II)
AI Agent (old §7)	Single service account	A **department of five named agents** — see Part II
CRM (old §9)	Recommend HubSpot	**Built in-house** after Tier 1 + 2; no external vendor changed
Admin (old §6.9)	Module-only roles	**Unified admin** governing both the SaaS and the future CRM
Database	"DDL deferred to the developer"	**Built and live** — this spec is written to it (Appendix B)
Non-functional (old §12)	Targets only	All **active**, delivered in Tier 3

BINDING VS ILLUSTRATIVE

As in v1.0: **field lists, behaviours, compliance rules, and the agent memory protocol are binding.** The mock-ups throughout — colours, dimensions, exact layout — are illustrative and open to change during build.

Contents

1 Executive summary

CarboNature is building a **verified-carbon factory**: a system that turns agricultural land management into high-integrity Verra Verified Carbon Units (VCUs), and does it faster each cycle. It is delivered in two tiers on one shared database.

Tier 1 — the AI-driven MRV module is the operating layer for the measurement, reporting and verification work itself. It manages soil-sampling cycles on a Mapbox map, issues work orders to field contractors over the Model Context Protocol (MCP), ingests the laboratory's SOC datasheet, runs the DNDC and DayCent carbon models, computes GHG emissions from fuel and nitrogen fertiliser through the VM0042 GHG Calculator, and scores every cycle against the VM0042 v2.2 hard checks. It supports all three quantification approaches — **QA1 (Measure & Model), QA2 (Measure & Remeasure), and QA3 (default factors)** — and is built for the ICVCM Core Carbon Principles (CCP) label.

Tier 2 — the Verified Credits Factory is an autonomous AI-agent department that operates the whole pipeline. Five named agents — **John** (head), **Reveka** (validation), **Dave** (verification / AI-MRV), **Ron** (marketing), and **Jennifer** (admin) — each carry a role prompt, a set of connected tools, and a growing library of skills, coordinated through a control-tower dashboard that shows, per project, how close each project is to issuance and how many credits the next Verification round will yield. Each agent keeps **long-term memory** in the project database so it improves cycle over cycle.

The whole system runs, for now, on the two **demo farms** — Elad Farm (Kenya) and Nitzan Veg-Tech Farm (Israel) — integrating with the CarboNature SaaS: pulling each farm's plots and Project Activities from the marketplace, and pushing forecast credits and readiness back into the farmer and credit-buyer dashboards. A dedicated **CRM is built in-house** after Tiers 1 and 2; no external CRM is procured.

THE NORTH STAR

The AI-Agent Department's mission, and the metric everything is judged against: **maximise the volume of Verra-verified carbon credits generated, in the shortest possible time.**

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2 Tier roadmap & priority backlog

Development proceeds tier by tier. Tier 1 ships the MRV module end-to-end on the demo farms; the moment it is accepted, Tier 2 (the agent department) begins; Tier 3 is hardening and scale. Within Tier 1 the priority tags P1 P2 P3 order the screens.

#	Capability	Tier	Notes
1	SSO login + workspace	T1 · P1	Single-tenant to start; project-scoped from day one
2	Project Map (Mapbox) — plots, strata, points, BSL	T1 · P1	Reads plot geometry from the live database / SaaS
3	Plot Details (history, lifecycle, lab, model runs)	T1 · P1	Pulls Project Activities from the marketplace pop-up
4	Sampling Plan Generator — 3-year view, ≥5 composites	T1 · P1	Grouped by project, filterable by farm
5	Work Order (PDF + email + MCP token, 14-day window)	T1 · P1	Starts real campaigns with the sampling lab
6	MCP Sampler view (GPS + barcode + photo)	T1 · P1	Field capture over the Model Context Protocol
7	Excel ingestion — SOC datasheet v2.0	T1 · P1	DIN 19539 fractions, ESM inputs, dry combustion
8	GHG Calculator engine (QA3)	T1 · P1	Fuel CO_2 + fertiliser N_2O; matches the workbook exactly
9	Model Run Console — DNDC v9.x / DayCent v6.x	T1 · P2	Paired baseline+project, Monte Carlo uncertainty
10	Compliance score — VM0042 hard checks + CCP	T1 · P2	Live in the DB as `evaluate_compliance()`
11	MVR + IME workflow (QA1)	T1 · P3	VMD0053; IME contracted by the VVB
12	Unified Admin / permissions (SaaS + CRM)	T1 · P1	One admin governs both systems
13	**Verified Credits Factory** — 5 agents + dashboard	Tier 2	Part II of this document
14	Agent long-term memory protocol	Tier 2	Postgres + pgvector, email-connected
15	In-house CRM · unified admin · NFR hardening	Tier 3	Built after Tiers 1 & 2

PART I

Tier 1 — the AI-driven MRV module

The operating layer for one measurement-reporting-verification pipeline: sampling, lab ingestion, carbon modelling, GHG accounting, and VM0042 v2.2 compliance — running on the database that is already live.

3 Scope & methodology alignment

The module is in **full alignment with Verra VM0042 v2.2** (Improved Agricultural Land Management, active 21 Oct 2025) and is built to qualify for the **ICVCM Core Carbon Principles (CCP)** high-integrity label. It supports all three quantification approaches simultaneously, because a single grouped project can carry plots on different approaches:

Approach	What it does	Where it lives in the module
QA1 — Measure & Model	DNDC / DayCent simulate SOC stock change against measured true-up points	Model Run Console · MVR / IME workflow
QA2 — Measure & Remeasure	Direct SOC remeasurement against ≥3 baseline control sites	Sampling lifecycle · ESM stock computation
QA3 — Default factors	Fuel CO_2 and fertiliser N_2O from IPCC 2019 emission factors	GHG Calculator engine (fully built & verified)

Cycle timing

The interval between sampling rounds is driven by the crop cycle, and **can be as long as 10 months**. In short-growth crops a further sampling round may occur in **under a year**. The sampling-cycle state machine and the two trigger types (end-of-growth-cycle, and a cap from the previous cycle) accommodate both without assuming an annual cadence.

The two workbooks the module operates against

The module works in lockstep with the two spreadsheets attached to this spec, both already reconciled with the database:

- **CarboNature_SOC_Datasheet_v2.0** — the file that goes to the laboratory, where SOC and bulk-density results are filled in. It carries the DIN 19539 carbon fractions (TOC = TOC400 + ROC600, TIC subtracted), ISO 11272 bulk density, coarse-fragment columns, and the dry-sample-mass + probe-area columns that make Equivalent-Soil-Mass reporting possible.

- **GHG_Calculator_VM0042_v2.2_OpenField_v1** — monitors emissions from nitrogen-based fertiliser use and fuel consumption in the farm operation. The module's emission engine reproduces this workbook to four decimals (Appendix B).

Model validation & the VVB

For the QA1 models, the **VVB appoints its own expert to audit the model calibration**. The module's **Model Validation Report (MVR)** and the **Independent Modeling Expert (IME)** workflow are built so that Dave (the Verification agent) communicates with that VVB-appointed expert directly. Per VMD0053, **the IME is contracted by the VVB, not by CarboNature** — the database records this explicitly.

> CONFIRMED – MONTE CARLO UNCERTAINTY
>
> **Yes: VM0042 v2.2 supports Monte Carlo uncertainty.** Its §8.6.1.2 defines the Monte Carlo propagation path (Equations 65–69) for QA1, sampling the posterior predictive distribution of a Bayesian-calibrated model, and **recommends L = 500–1000 iterations**. It sits alongside the analytical error-propagation path (Eqs 60–64); either feeds the Equation 74 uncertainty deduction. Dave's Uncertainty-Deduction skill uses the Monte Carlo route, VMD0053-aligned.

4 Personas & roles

Tier 1 has one human super-admin, one AI operator, and two field roles. Note the shift from v1.0: the "AI-MRV Agent" is no longer an anonymous service account — it is **Dave**, the Verification Manager agent from Tier 2, operating the module.

Persona	Type	Responsibility in the module
Super Admin — Nitzan	Human · full access	Creates projects, approves baselines, signs off MVRs with the IME, manages tokens and model licences, holds override authority on every agent decision
Dave — AI-MRV Agent	AI · service identity	Runs the whole verification pipeline: proposes sampling plans, runs DNDC/DayCent, ingests lab data, computes the GHG Calculator, drives compliance. Per-action AUTO / CONFIRM / OFF policy (default CONFIRM for anything that creates an artifact)
MRV Technician	Human · reports to Dave	**Not an approver.** Executes sampling rounds on the ground with the lab's sampling teams and supervises adherence to CarboNature's protocols and Excels; feeds lab results back on the SOC Datasheet v2.0; collects fuel and fertiliser data from farmers for the GHG Calculator, after Dave emails follow-ups flagging the missing items
Soil Sampler	External contractor · MCP token	Reaches the module via a work-order-scoped MCP token; captures GPS, barcode, photo and notes per point. Cannot edit the plan or see other projects or lab results

REPORTING LINE

The MRV Technician's direct manager is **Dave** (the AI Verification Manager); above both sits the human **Super Admin & CEO (Nitzan)**. Approval of Dave's proposals — sampling plans, work orders, model runs — rests with Dave and the Super Admin, never with the Technician.

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5 Architecture — on the live database

Unlike v1.0, which deferred the data layer, the module is now written to a **database that already exists and is verified**. The DDL is not a future exercise; it is 16 applied migrations across seven build stages. The architecture below reflects what is running.

Layer	Choice	Status
Database	PostgreSQL 16 · PostGIS 3.4 · pgvector — schema `mrv`, 41 tables, 4 views	`live on AWS RDS eu-west-1`
Spatial	All geometry `SRID 4326`, GIST-indexed; point-in-plot < 100 ms	`built`
Evidence integrity	Append-only triggers on every evidentiary table; full audit log	`built`
Object storage	S3 — raw lab workbooks, photos, model outputs, MVRs · KMS · Glacier 10 yr	`provisioned`
GHG engine	`mrv.compute_emissions()` — reproduces the workbook exactly	`built & verified`
Compliance engine	`mrv.evaluate_compliance()` — VM0042 hard checks, 0–100 score	`built`
Agent memory	`mrv.agent_memory` — pgvector(1536), per-agent, per-project	`table live`
Model runners	Containerised DNDC v9.x / DayCent v6.x (ECS/Fargate + NAT)	`Tier 2 infra`
Web client	React + TypeScript · Mapbox GL JS	`Tier 1 build`
Agent runtime	Claude (Anthropic) + MCP tool servers	`Tier 2 build`

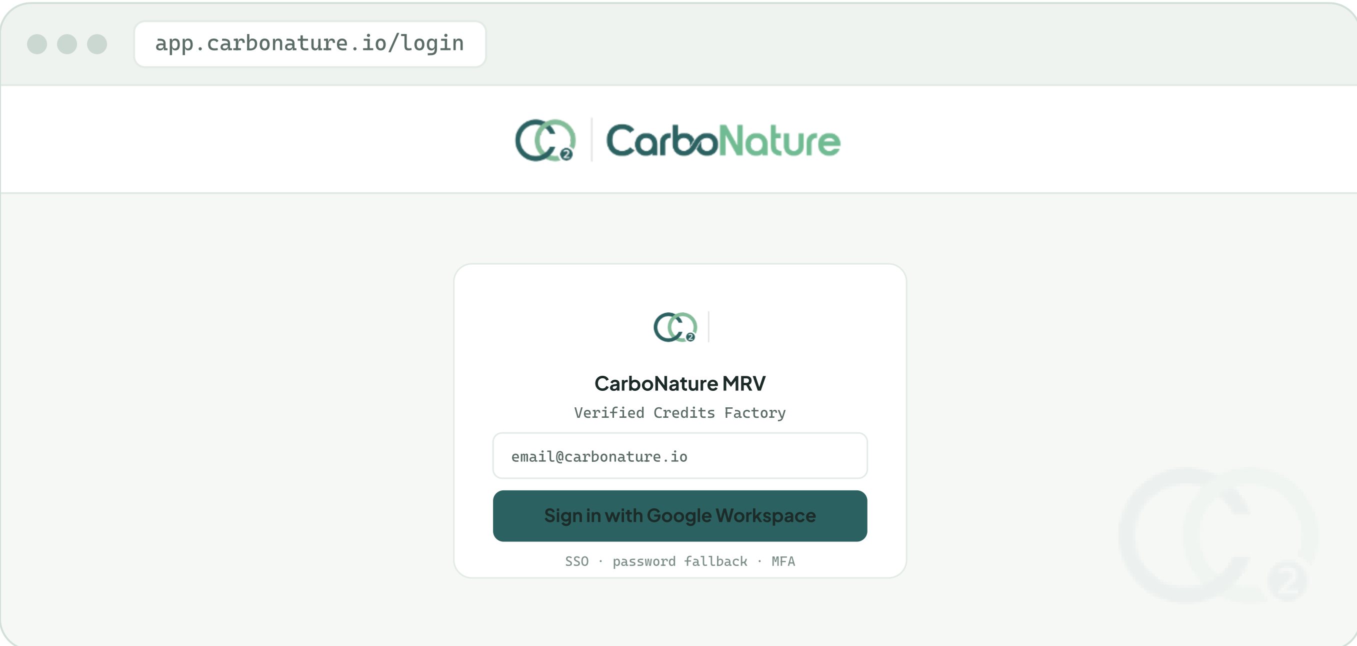
The full data-flow and system-architecture diagrams accompany this spec as a separate two-page PDF. Appendix B summarises how each screen maps onto the database.

6 The eight screens

v1.0 had nine screens. The old §6.8 "Dashboard — Forecast vs Actuals" is **removed**: its job — readiness and forecast — is done far better by the Tier-2 Agents Dashboard (Part II), which also feeds those figures back to the SaaS. The remaining eight screens are updated below; each carries a mock-up. Field lists and validation are binding; the mock-ups are illustrative.

6.1 Login & Workspace `P1` `/login · /workspace`

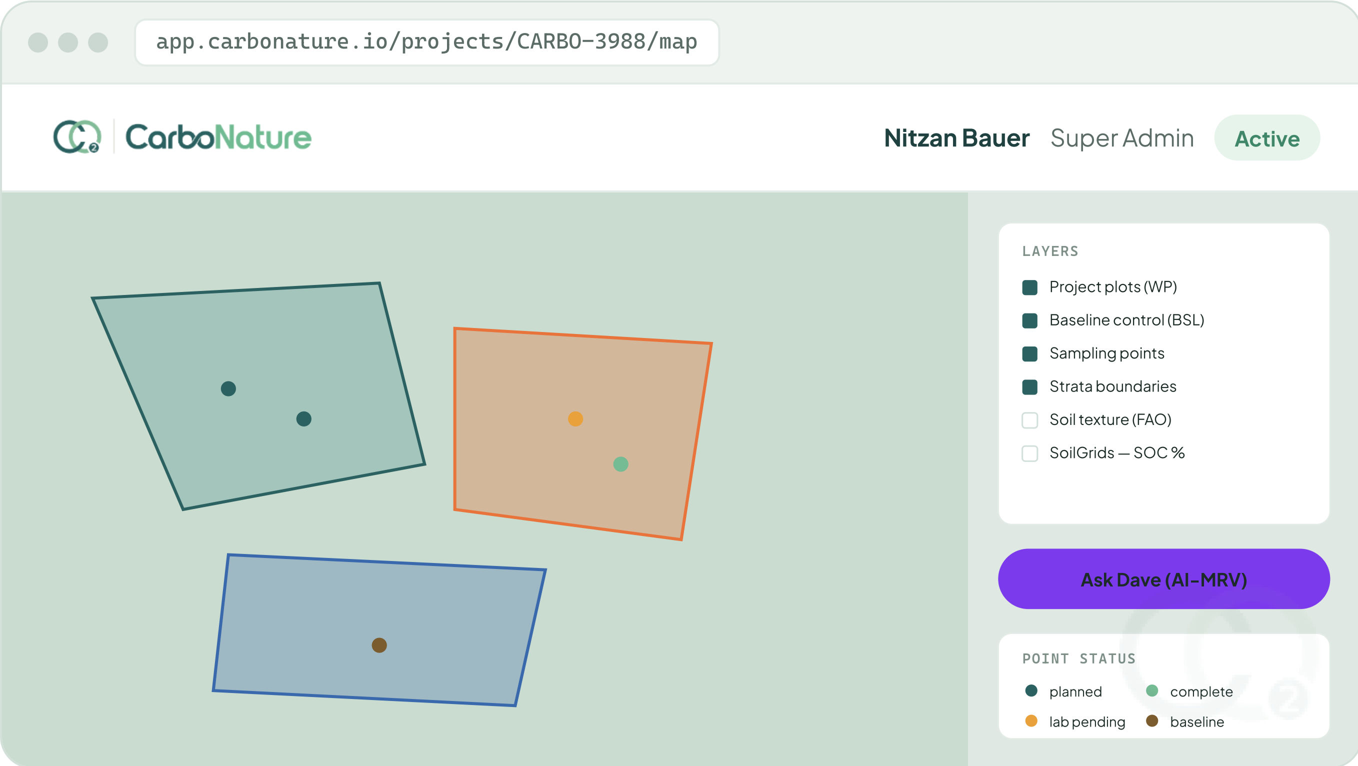
Single entry point for internal users via Google Workspace SSO; external samplers never log in here — they use a one-tap MCP token. Three failed passwords → 5-minute lockout; SSO preferred for `@carbonature.io`. First login routes to Project Map. **Unified sign-on:** the same identity carries into the future CRM and the SaaS admin (§6.8).



Screen 1 · Login — SSO first; one identity across the module, the SaaS admin, and the CRM.

6.2 Project Map `P1` `/projects/{id}/map`

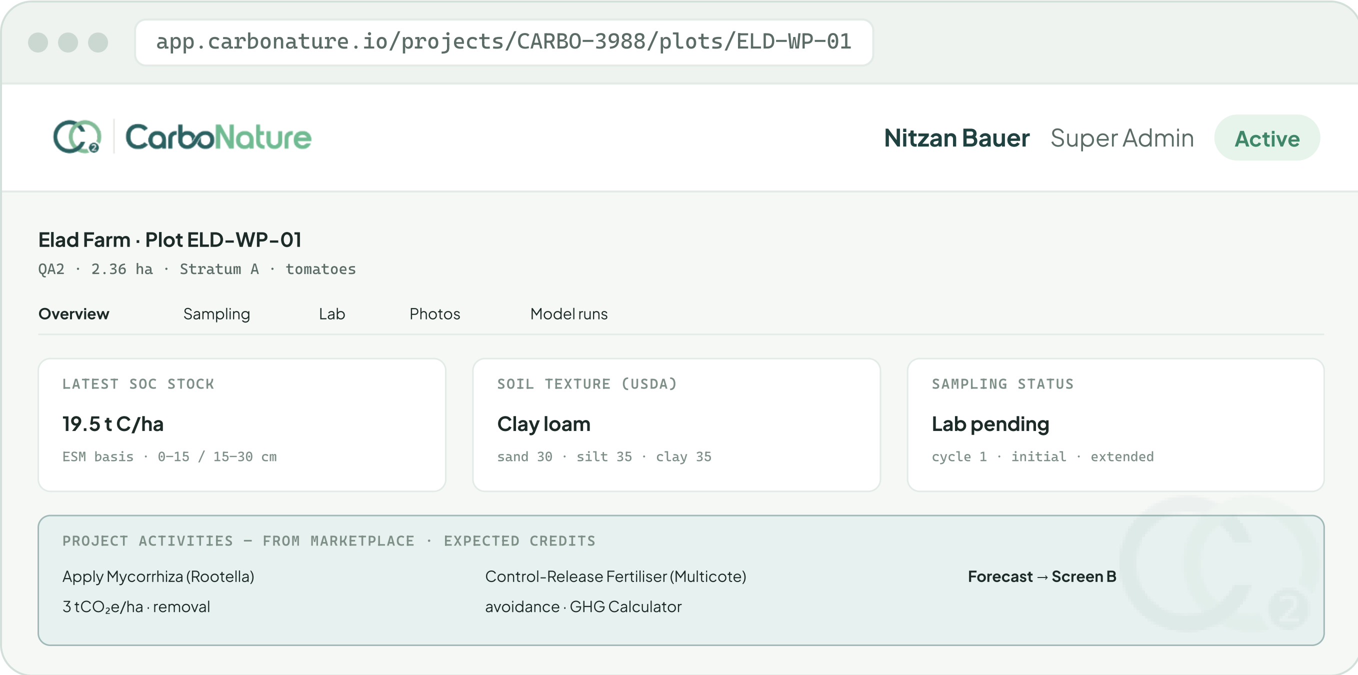
The primary spatial workspace on Mapbox: WP polygons, baseline control sites, sampling points colour-coded by status, and strata boundaries. Plot geometry is read from the live database, which is populated from the SaaS marketplace — so the map matches exactly what a farmer and a credit buyer see. Clicking a plot opens its Project-Activities pop-up (mycorrhizae, CoteN, mulching) and Plot Details. Reviewed against v1.0; layers and interactions unchanged except the geometry source is now the built database.



Screen 2 · Project Map — WP plots, strata, points by status, BSL; the "Ask AI" button now opens Dave.

6.3 Plot Details `P1` `/projects/{id}/plots/{plotId}`

One pane per plot, tabbed: Overview, Sampling history, Lab results, Photos & barcodes, Model runs. The **Project Activities** for the plot are pulled from the marketplace pop-up and shown here, so the expected-credit forecast can be built alongside the GHG Calculator (avoidance + removal). Reviewed against v1.0; adjusted only to source Project Activities from the marketplace and to add the credit-forecast link.

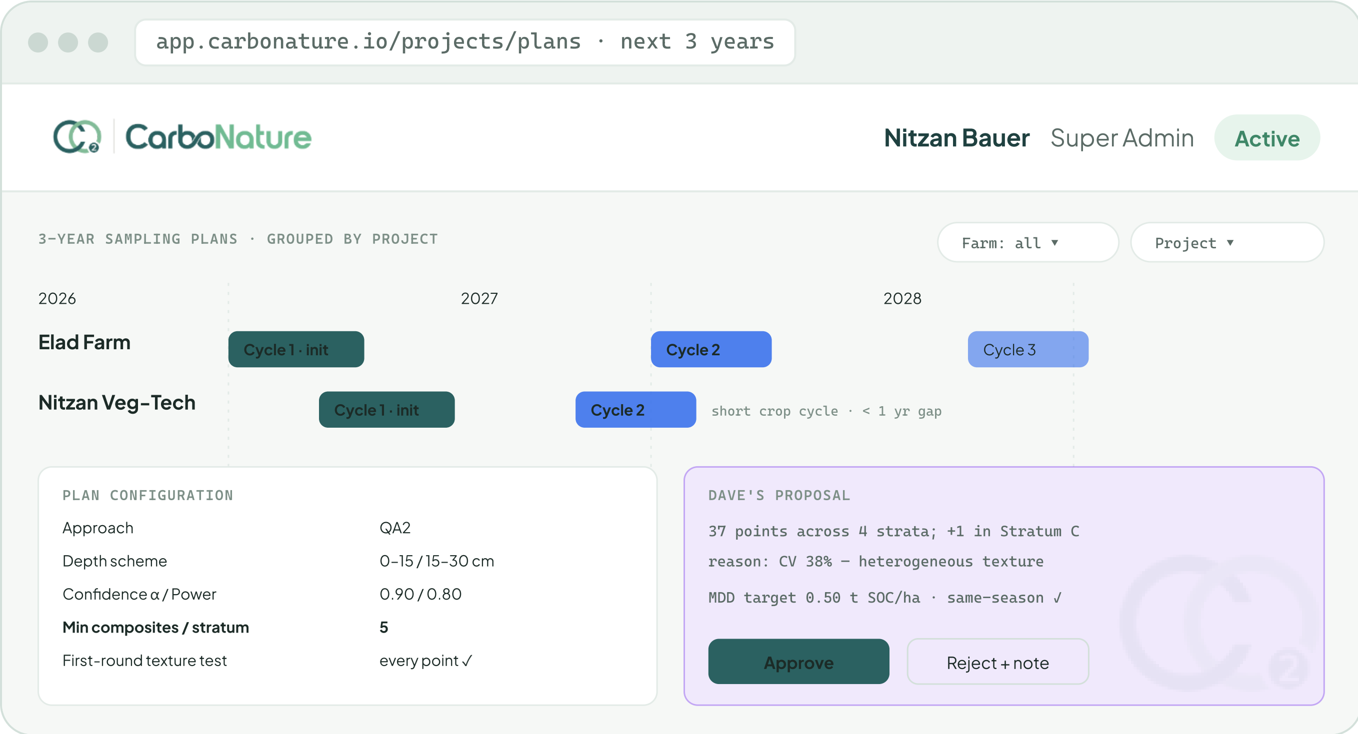


Screen 3 · Plot Details — SOC, texture, status, and the marketplace Project Activities driving the credit forecast.

6.4 Sampling Plan Generator P1 `/projects/{id}/plans`

The flagship planning screen. The user sees **all sampling plans for the coming 3 years, grouped by project and filterable by farmer/farm**; clicking a plan opens a detail window with its full parameters. The generator proposes a stratified-random plan that strictly meets VM0042 v2.2, and Dave overlays a recommendation with reasoning.

> **CHANGED FROM V1.0**
>
> Minimum composites per stratum is **≥ 5** (was ≥ 3). The generator's `MIN_COMPOSITES` hard check and the compliance engine both enforce 5. First-cycle behaviour still adds extra soil-composition samples to define the baseline, and **every point in the first round also receives a Soil-Texture test** whose sand/silt/clay result drives the stratification.



Screen 4 · Sampling Plan Generator — 3-year view grouped by project, ≥5 composites/stratum, first-round texture at every point.

6.5 Work Order P1 /projects/{id}/work-orders/{woId}

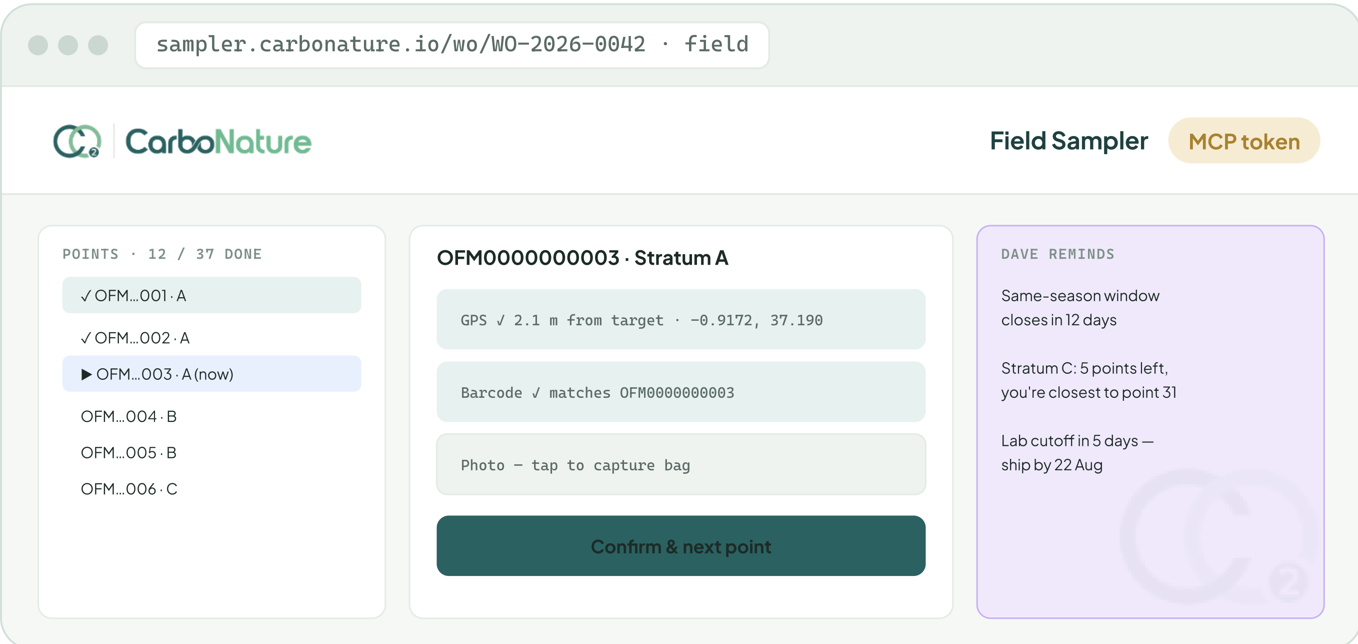
The bridge from a planned cycle to a contractor in the field: a printable PDF, an email, and a one-tap MCP activation token. The document carries the header, assignment (contractor, sampling window, lab destination, project lead), the sampling-points table (Sample ID, stratum, WP/BSL, lat/lon, depth, composite cores, re-visit flag), the VM0042 §8.2.1.3 field protocol, and the MCP connection block. States: Draft → Sent → In progress → Completed → Closed; each transition is written to the audit log.

> CHANGED FROM V1.0
>
> The MCP token now expires **14 days after the end of the sampling window** (was 7), and the window is **configurable** per work order.

Screen 5 · Work Order — PDF + email + scoped MCP token, expiring 14 days after the (configurable) window.

6.6 MCP Sampler View P1 `/sampler/wo/{woId}?token=…`

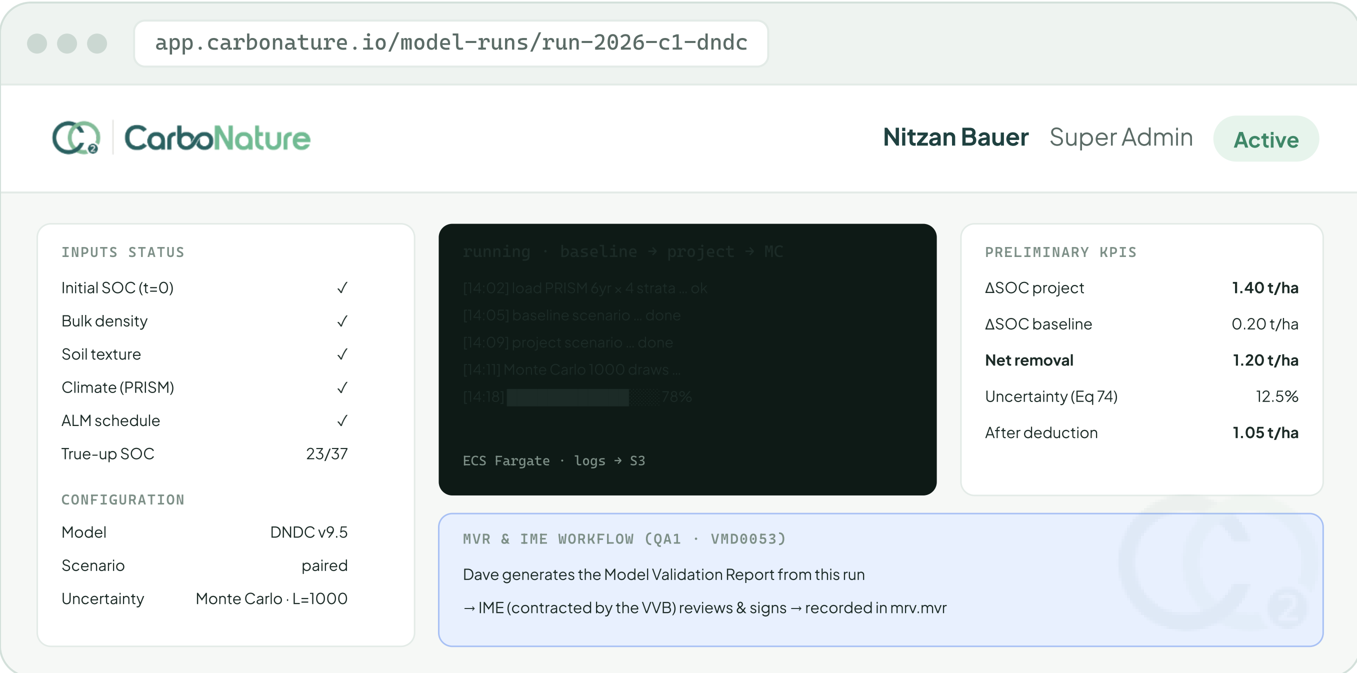
The field web view an external contractor opens from the token, connected to Dave through MCP. Each capture — GPS, barcode scan, photo — is an MCP tool the agent can also call and inspect, and Dave surfaces context-aware reminders ("same-season window closes in 12 days"). Validation per point: GPS within 25 m of target (soft), barcode must match the expected Sample ID exactly (hard), photo required, then "Confirm & next point". Reviewed against v1.0; unchanged except the token window (§6.5) and that the agent is now Dave.



Screen 6 · MCP Sampler View — GPS + barcode + photo per point, with Dave's field reminders over MCP.

6.7 Model Run Console — DNDC / DayCent P2 `/projects/{id}/model-runs/{runId}`

The control surface for QA1. Inputs are pre-validated (initial SOC, bulk density, texture, PRISM daily climate, the per-plot ALM schedule, true-up SOC); the run executes in a containerised worker; logs stream live; the project-vs-baseline SOC trajectory and preliminary KPIs render as it runs. Results write back to the database (`model_runs`, `model_results`) and S3. **Monte Carlo uncertainty** (L 500–1000, VMD0053) produces the Equation-74 deduction. Reviewed against v1.0; adjusted so results land in the built tables and the MVR/IME workflow is explicit.

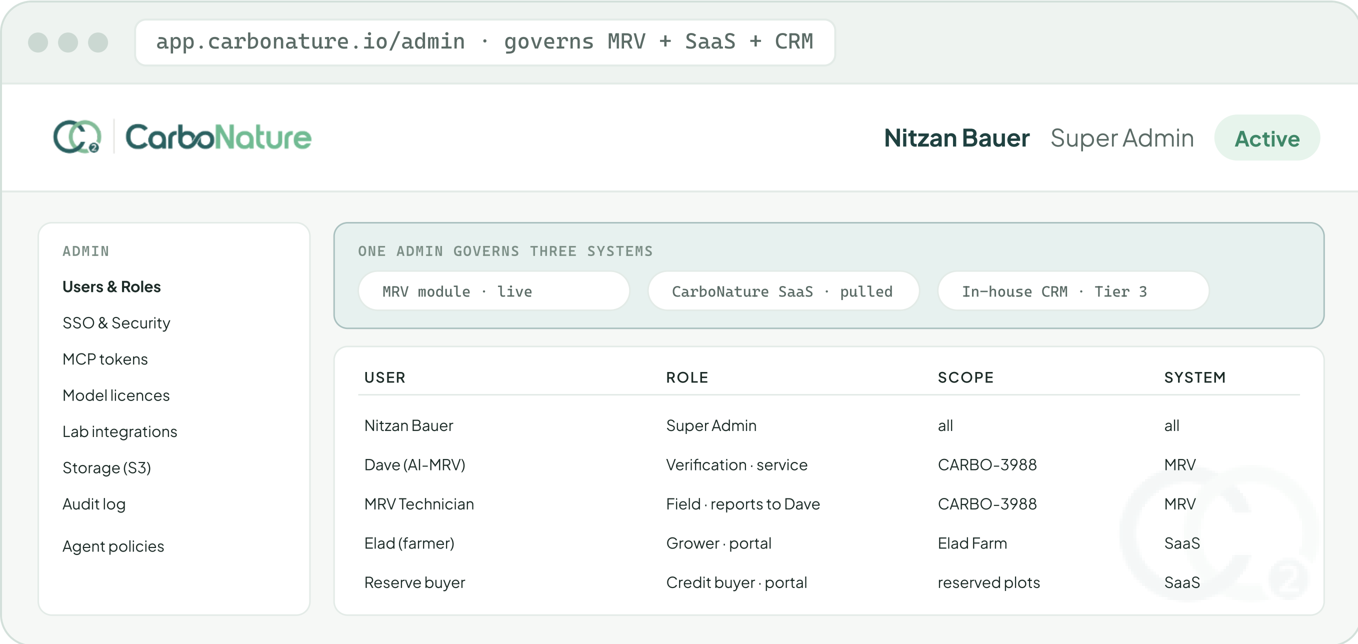


Screen 7 · Model Run Console — DNDC/DayCent paired run, Monte Carlo deduction, MVR handed to the VVB's IME.

6.8 Admin / User Management — unified P1 `/admin`

A **single, unified permissions-and-admin system that centralises administration for both the MRV module / future CRM and the CarboNature SaaS**. The module connects to the SaaS admin, pulls all of its data, and governs it from here — one place to manage users, roles, project access, SSO, MCP tokens, model licences, storage, audit log, and lab integrations, across both systems.

> CHANGED FROM V1.0
>
> Admin is no longer module-only. It is the **authoritative admin for the SaaS as well**, so a user, role, or permission is defined once and applies everywhere. The in-house CRM (Tier 3) plugs into the same system.



Screen 8 · Unified Admin — one permissions system governing the MRV module, the SaaS, and the future CRM.

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7 VM0042 v2.2 + CCP compliance hooks

Compliance is not a report generated at the end; it is computed continuously by `mrv.evaluate_compliance()` in the database, which scores every farm-cycle 0–100. A hard failure caps the score below 100 regardless of warnings, so the dashboard reads red the moment a rule is broken. The module must qualify for both **VM0042 v2.2** and the **ICVCM CCP** label.

#	Hard check (must pass)	VM0042	Status
1	Stratified random sampling every cycle	§8.2.1.2	enforced
2	**≥ 5 composite samples per stratum** was 3	§8.2.1.2	enforced
3	ESM reporting — 2 depth increments (0–15 / 15–30 cm)	§8.2.1.6	enforced
4	Same-season sampling window respected	§8.2.1.1	enforced
5	QA2: ≥ 3 baseline control sites, each ≤ 250 km, all 9 similarity criteria	§8.3 · Table 7	enforced
6	QA1: model validated per VMD0053; MVR present and IME-signed	§8.6.1.3	P3
7	Lab ISO/IEC 17025 (or NAPT / GLOSOLAN)	§8.2.1.4	enforced
8	Dry combustion (Dumas) unless a deviation is documented	§8.2.1.4	enforced

Soft warnings (raise, don't block): per-stratum CV > 30% (suggest +1 sample next cycle); model true-up due within 6 months; verification approaching with > 5% sampling shortfall.

ICVCM CCP – THE DSM CONDITION

To carry the CCP label, SOC must be measured by any permitted technique **except Digital Soil Mapping (DSM / VT0014)**, which ICVCM did not assess. The module and Reveka's PDD infrastructure are built to that condition; VT0014 is deliberately out of scope for CCP-labelled projects.

8 Canonical SOC schema — from the SOC Datasheet v2.0

The laboratory returns `CarboNature_SOC_Datasheet_v2.0`; the ingestion pipeline maps it to the canonical `mrv.soc_measurements` table, recomputing SOC in the service layer rather than trusting the sheet's formula. The v2.0 datasheet reflects everything the module learned about dry combustion and ESM.

Field	Canonical column	Notes
Sample ID	`sample_id`	`OFM` + 10 digits, e.g. `OFM0000021615`
Depth increment	`depth_top / depth_base`	0–15 & 15–30 cm — two rows per sample (ESM)
TOC 400 / ROC 600 / TIC 900	`toc_400 · roc_600 · tic_900`	DIN 19539 / ISO 17505 ramped fractions
TOC (organic)	`toc_pct` **(generated)**	**TOC = TOC400 + ROC600**; never entered by hand
Bulk density	`bulk_density`	ISO 11272, fine-earth (< 2 mm) only
Coarse fragments	`small_cf · large_cf`	2–10 mm mass % · > 10 mm vol %
Dry sample mass · probe area	`dry_sample_mass_g · probe_area_cm2`	**Required for ESM** — soil mass = mass ÷ area (§8.2.1.6)
SOC stock	`soc_t_per_ha`	Factor ×100 (not ×1000); recomputed in service layer
Texture (cycle 1)	`sand · silt · clay → usda_class`	Drives stratification; sum = 100

TWO CORRECTIONS BAKED INTO V2.0

TOC is not TOC400 alone — treating the labile fraction as total organic carbon under-reports any soil holding char or soot (common where residue is burnt). And **ESM needs mass and area, not bulk density and depth** — omitting the dry-mass and probe-area columns makes equivalent-soil-mass reporting impossible in retrospect, which is irreversible.

9 MCP sampler integration

The external contractor has no SSO account. When Dave (or the Super Admin) sends a work order, the system issues a scoped MCP token; the contractor opens the URL in any browser and the Sampler View (Screen 6) speaks to the Sampler MCP server. The same tool surface — `capture_gps`, `scan_barcode`, `capture_photo`, `submit_point` — is used by both the human contractor and Dave, so there is no code drift between the human and agent paths.

- **Scope:** one work order and its sampling points only.
- **Expiry:** end of the sampling window **+ 14 days**, configurable `changed`.
- **Revocable** by Super Admin or Dave at any time; stored as a hash in `mrv.mcp_tokens`, never the raw token.

10 Acceptance criteria — Tier 1 MVP

1. Super Admin logs in via SSO and sees the demo project(s) on the map.
2. Any plot opens a populated Plot Details panel, including its marketplace Project Activities.
3. A 3-year sampling plan can be generated (manual rules), grouped by project and filtered by farm, meeting **≥ 5 composites/stratum**, and approved.
4. Approving a plan creates a Work Order with a downloadable PDF, a contractor email, and an MCP token URL valid for the window + 14 days.
5. An external sampler opens the token URL and captures GPS + barcode + photo + notes for at least one point.
6. The `CarboNature_SOC_Datasheet_v2.0` lands and within a minute the parsed rows appear under the plot's Lab tab and the map flips the point green.
7. The GHG Calculator engine reproduces a baseline farm-year to four decimals (fuel CO_2 + fertiliser N_2O).
8. The compliance score computes for a cycle, flagging any hard-check failure.
9. The unified Admin shows users from both the MRV module and the SaaS, governed from one place.
10. Every action above is written to the audit log with timestamp, actor, action, and target.

All runs are on the designated demo farms — **Elad Farm** and **Nitzan Veg-Tech Farm**.

Verified Credits Factory — the AI-Agent Department

An autonomous department of five AI agents that operates the whole credit pipeline — validation, verification, marketing, and admin — coordinated through a control-tower dashboard, each agent improving cycle over cycle through long-term memory.

11 Department mission & overview

The AI-Agent Department's mission is the company's north star: **generate the maximum volume of Verra-verified carbon credits, in the shortest possible time.** Every agent, skill, and screen is measured against that one outcome.

The department is a hierarchy: **John** heads it and owns the dashboard; beneath him, **Reveka** drives validation, **Dave** drives verification (he is the AI-MRV agent that operates the Tier-1 module), **Ron** drives marketing, and **Jennifer** keeps admin and records clean. A human **MRV Technician** executes field work under Dave. Everything reports, ultimately, to the CEO.

Agent	Title	Owns
John	Head of Department	The mission, the four agents, and the control-tower dashboard
Reveka	Validation Manager	PDD, baseline, additionality, VVB validation audit, grouped-project design
Dave	Verification Manager (AI-MRV)	Sampling → modelling → GHG accounting → VVB verification → VCU issuance
Ron	Marketing Manager	Farmer acquisition (Funnel A) and credit-buyer acquisition (Funnel B)
Jennifer	Admin Secretary	Document centralisation, CRM/DB hygiene, scheduling, board-meeting protocol

12 The five agents

Each agent carries a role prompt, a set of connected tools, and a library of skills — some shared, some the agent develops itself to reach its goal faster. Below is each agent's mission and the core of its responsibilities; the full role prompts are stored in the agent registry and shown verbatim in the dashboard's agent modal (§13, Screen A).

John — Head of Department

Mission: generate the maximum volume of Verra-verified credits in the shortest possible time.
Reports to: the CEO (Nitzan).

- **Lead the four agents** — direct, prioritise and coordinate Validation, Verification, Marketing and Admin so they pull in one direction.
- **Detect and clear bottlenecks** in the validation/verification pipeline; master the Verra and VVB process and its common failure points; benchmark comparable VM0042 / ALM projects to extract the levers that shorten time-to-issuance.
- **Maximise credit yield** — command both Carbon Removal and Carbon Avoidance generation, and drive high-yield activities (mycorrhizae, CoteN, mulching) across more farmer hectares.
- **Own and operate the Agents Dashboard** — feed it complete, current data from every source (Verra registry, the database, the model engine, each agent), run it day to day, consolidate the full picture, and govern the outbound feeds to the SaaS so no numbers diverge across audiences.
- **Controls & QA** — reconcile forecast-vs-actual, validate the credit-allocation math against the model and stakeholder agreements, and confirm figures are audit-ready before they surface.
- **Ensure integrity & compliance** — keep every project aligned with VM0042 v2.2 and the ICVCM CCP standard; report to the CEO on volume, cycle time, pipeline health, and blockers.

Reveka — Validation Manager

Mission: complete the validation certification to the highest standard in the shortest time.
Reports to: John.

- **Produce a submission-ready PDD in ≤ 30 minutes** from Verra's official Word template, in full alignment with VM0042 v2.2. Run a preliminary digital questionnaire — **directed to the Super Admin (Nitzan), not the client** — to gather inputs; on the first round, partially complete the PDD and submit under "Under Development" to register the project.
- **Own the single source of truth for project data** (jointly with Jennifer) — farmer records, plot geometries, activities, evidence — and run QA/QC on boundaries, areas and soil inputs before every submission.
- **Document baseline, additionality, applicability** exactly as VM0042 v2.2 requires; compile the eligibility evidence pack linking each activity to its VM0042 reference (Appendix 1 p.141; definitions p.9).
- **Design the project as a grouped project** so farmers and instances can be added post-validation; prepare the KMZ/KML for every farmer's plots and the Listing Representation form.

- **Ensure the monitoring plan meets the CCP condition** — SOC by any permitted technique **except DSM**. Build the PDD-level infrastructure for the DayCent / DNDC models and for the ICVCM CCP label.
- **Manage all VVB validation communications**, resolve every reject/finding, run the public-comment step, and hand off cleanly to Dave once validation is complete.
- **Skill — PDD Generator:** a dedicated skill that researches every issued PDD in the category on the Verra registry, anchors the draft to the closest projects (crop, geography, climate, activities, approach), and writes the PDD end-to-end from the official template, traceable to its sources.

Dave — Verification Manager (AI-MRV)

Mission: complete verification — from field measurement through carbon modelling to VVB verification and VCU issuance — to the highest standard in the shortest time. **Reports to:** John. **Operates the entire Tier-1 module.**

- **First-round texture at every point** — ensure every sampling point in the first round gets a soil-texture test, and use the results to divide the project into homogeneous strata; apply the same texture-based stratification to the baseline (BSL) plots.
- **Run a dedicated skill suite:** Stratification (VM0042 §8.2.1, re-stratify as texture/lab data arrive), Baseline-definition (BSL sites: QA2 ≥3 within 250 km, all 9 Table-7 criteria), DNDC (v9.x, paired, stratum-by-stratum, true-up), DayCent (v6.x, same wrapper), Uncertainty-Deduction (Monte Carlo, VMD0053, L≈500–1000), and a GHG-Calculator skill that runs the whole workbook end-to-end.
- **Own the two VM0042 audit workbooks** — and above all **understand deeply what each one computes and why**, not merely how to fill it:
    - **GHG Calculator** (Approach 3): baseline (3-yr pre-project average) and project activity per farm-year — fertilisers, fuel, residues, N-fixing crops — commanding direct and indirect N_2O, N_2O from N-fixing residues and residue burning, CO_2 from fuel, the conservativeness auto-selection of factors, the QA3-vs-QA1 soil-N_2O double-counting switch, leakage, and the roll-up to estimated VCUs with AFOLU buffer withholding.
    - **SOC & Bulk-Density Datasheet v2.0**: Sample IDs, depth increments, coarse fragments, the DIN 19539 fractions (TOC = TOC400 + ROC600, TIC subtracted), ISO 11272 bulk density, and the dry-mass + probe-area columns for ESM (§8.2.1.6); enforce dry combustion, ISO 17025 lab consistency, and the built-in QC checks.
- **Defend every number** — trace any output back to its inputs, equations, and VM0042 / IPCC 2019 source, and answer the VVB's questions on both workbooks without hesitation.
- **Enforce the hard checks** (stratified random; **≥5 composites/stratum**; ESM ≥2 increments; same-season; QA2 ≥3 control sites ≤250 km; QA1 MVR + IME-signed; ISO 17025; dry combustion) and surface the soft warnings.

- **MVR & IME sign-off** for QA1 per VMD0053; interface with the VVB and its appointed modelist/auditor; resolve every reject until VCUs issue; keep a full replayable audit log of every run, input set, and parameter version.

Ron — Marketing Manager

Mission: drive the two conversions that feed the north star. **Reports to:** John.

- **Funnel A — farmer acquisition:** get as many growers as possible to join CarboNature (register on `app.carbonature.io` and enrol their plots). More enrolled hectares = more credits produced.
- **Funnel B — buyer acquisition:** get as many credit buyers as possible to close financing deals (Reserve → login → deal flow on `carbonature.io`). More funded deals = more credits sold and pre-financed.
- **Prospecting, multichannel outreach, LinkedIn campaigns, organic SEO** toward those two conversion actions; lead qualification and routing (hectares on the farmer side; mandate/volume/budget on the buyer side); nurture cadences so no lead goes cold.
- **Farmer onboarding via the `onboard-new-client` skill** — when an acquired farmer registers, create the farm's data-driven public page on `carbonature.io` and add its card to the Marketplace, turning a new farmer into a live, buyable listing.
- **Content & collateral** — decks, one-pagers, case studies (Groundwork BioAg, Agreena, AgriCarbon), SEO articles, LinkedIn content; A/B-test and optimise the AI-SDR campaigns for both funnels.

Jennifer — Admin Secretary

Mission: keep the department running smoothly — accurate intake, organised records, reliable scheduling, clean data. **Reports to:** John.

- **Document centralisation** — maintain each client's folders and centralise all project documents (registration PDFs, contracts, KML/KMZ, kickoff docs, deals tracker) in one organised, current store.
- **Database & CRM hygiene** — enter and maintain project records in the database and keep the CRM clean, deduplicated and current so every agent works from accurate data.
- **Scheduling, correspondence, records & retrieval** — calendars, meetings, the shared inbox routing, and versioned, easy-to-retrieve records for agents, the VVB, auditors and counsel.
- **Board-of-Directors meeting protocol** — run CarboNature's board management end-to-end as a scheduled protocol: notices, quorum, agenda and board pack, official documents, the signature workflow, minutes and resolutions, and a version-controlled corporate-records

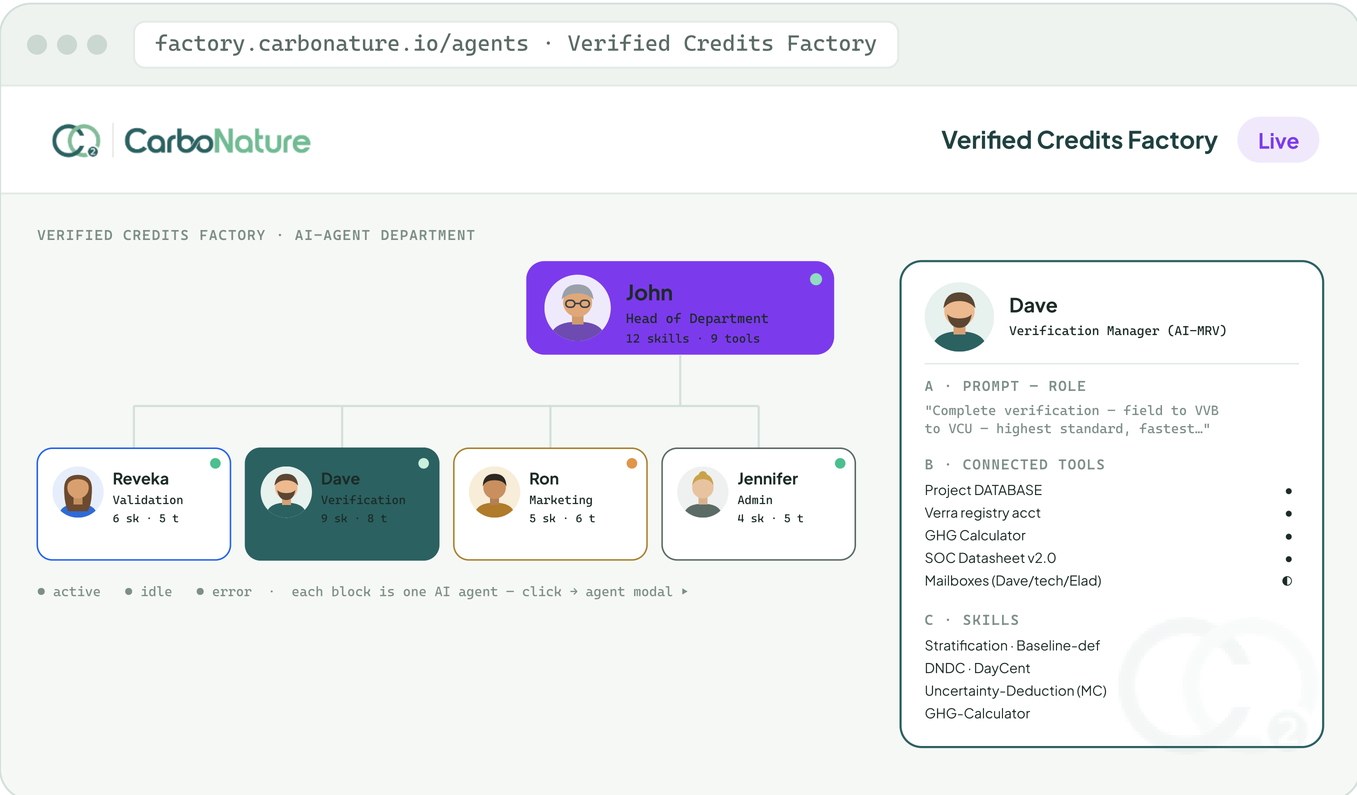
repository. Transition: this currently runs on a small "CarboNature Agent"; when Jennifer is established the task transfers to her and that agent is retired.

13 The Agents Dashboard — the control tower

The dashboard does two things at once: it visualises the department as an interactive hierarchy of clickable agents, and — for every project separately and at any moment — it shows how close the project is to credit issuance and forecasts the credits the next Verification round will yield, broken down per stratum and allocated across stakeholders. **Screens A and B are the MVP**; C and D follow once the forecast engine and stakeholder-agreement data are wired.

Screen A — Agent hierarchy MVP

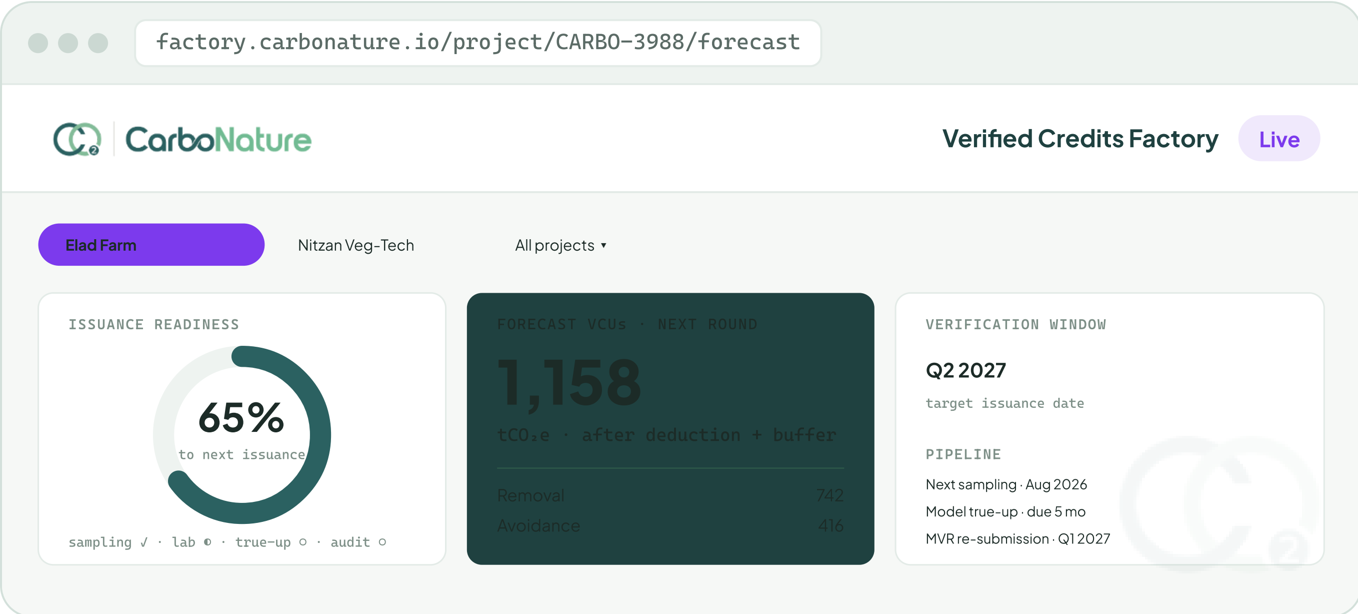
An org chart: John on top, Reveka, Dave, Ron and Jennifer beneath. Each block shows the agent's name, title, an avatar, a status dot (active / idle / error), and a badge with its skill and tool counts. Clicking a block opens a modal with three sections — **Prompt** (the full role definition, verbatim), **Connected tools** (mailboxes, the Verra registry account, the project database, the GHG Calculator, the SOC datasheet — each with a connection status), and **Skills** (shared plus agent-developed, e.g. Reveka's PDD Generator; Dave's DNDC, DayCent, Stratification, Uncertainty-Deduction skills).



Screen A · Agent hierarchy — clickable org chart; the modal shows each agent's prompt, connected tools, and skills.

Screen B — Issuance readiness & forecast `MVP`

Selecting a project scopes the whole dashboard to it. The KPI row answers, for that project alone: **how close are we, and how much will we get.** Readiness (%) is derived from the sampling-cycle state machine — sampling completion, lab results received, model true-up done, audit steps cleared. Forecast VCUs is computed from modelled net SOC removals plus quantified emission reductions, after the uncertainty deduction and buffer withholding — the same engine Dave runs. Both figures are live, per-project, with an all-projects roll-up.



`Screen B · Readiness & Forecast — per project: how close to issuance, and the forecast VCUs (removal + avoidance) for the next round.`

Screen C — Credit-issuance & allocation report `follow-on`

A drill-in per project. **Per-stratum forecast:** every stratum and the credits it will receive, split into Carbon Removal and Carbon Avoidance, with a stratum total and a project TOTAL. **Stakeholder allocation — the bottom line:** how the forecast credits are shared across credit buyers, farmers, and CarboNature, each tied to its indicative revenue, priced by credit type at current category rates. Exportable for the VVB and internal review. Split percentages and pricing are configurable per project and per grower agreement; values below are illustrative.

factory.carbonature.io/project/CARBO-3988/report

CarboNature

Verified Credits Factory Live

Elad Farm · next issuance · detailed report

PER-STRATUM FORECAST (TCO_2E)

STRATUM	AREA HA	REMOVAL	AVOIDANCE	TOTAL
A — clay loam	18.4	288	160	**448**
B — loam	14.1	214	128	**342**
C — sandy loam	9.2	156	96	**252**
BSL — control	3.3	—	32	**116**
TOTAL		742	416	1,158

WHERE CREDITS GO

Credit Buyers
579 VCU · $28,950

Farmers (50%)
347 VCU · $17,370

CarboNature
232 VCU · $11,600

priced by type · category rate
export → VVB / internal

Screen C · Allocation report — credits per stratum (removal vs avoidance) and the stakeholder split with revenue.

Screen D — Category market tracker `follow-on`

The dashboard follows verified-credit transactions in the same category (VM0042 / ALM soil-carbon) — recent deals, volumes, buyers, prices — to price the next issuance and keep monitoring going forward. Maintained by Ron; it feeds the pricing used in the Screen-C revenue figures.

factory.carbonature.io/market · VM0042 / ALM soil-carbon

CarboNature

Verified Credits Factory Live

Category market — VM0042 / ALM soil-carbon

RECENT VERIFIED DEALS

BUYER	VOLUME	$/T	TYPE
Microsoft	40,000	$58	removal
Agreena buyer	12,500	$47	avoid
Boomitra offtake	8,200	$51	removal
Regional buyer	5,000	$44	avoid

Screen D · Market tracker — category deals and prices (maintained by Ron) that set the Screen-C revenue figures.

14 Outbound feeds to the SaaS dashboards

Beyond its own screens, the Agents Dashboard is a **data source** that feeds the farmer and credit-buyer dashboards already built in the CarboNature SaaS — pushing the relevant figures into the relevant parts of each, so every audience sees one consistent source of truth. The same forecast engine and allocation logic drive the internal dashboard and both external feeds, so numbers never diverge.

Destination	Receives (relevant slice only)
Farmer dashboard	Their plots' issuance readiness, forecast credits for the next round, per-stratum breakdown for their land, and their credit/revenue share
Credit-buyer dashboard	Forecast available volume, issuance timing, the credits reserved/contracted to them, and matching pricing from the category market tracker

Each external dashboard receives only its own slice — a farmer sees their farm, a buyer sees their reserved credits. The full cross-stakeholder view stays internal to the Agents Dashboard.

15 Agent long-term memory protocol

A standalone protocol, implemented alongside the agents (independently of the dashboard build), that governs how every agent builds and manages memory. The long-term tier is what lets an agent improve cycle over cycle.

A · Two memory tiers

- **Short-term (session):** the working context of the current task; discarded or summarised at session end.
- **Long-term (persistent):** per-agent, per-project knowledge that survives across sessions — stored in the project database (`mrv.agent_memory`, PostgreSQL + pgvector) for semantic retrieval.

B · What each agent writes

Category	Example
Decisions & rationale	A stratification choice, a chosen baseline, a model parameter set — with the reasoning
Artifacts & versions	PDD versions, model runs and inputs, calculator outputs, submitted forms — versioned and traceable
VVB interactions	Rejects/findings and how they were resolved, so the same issue is never repeated
Parameters & references	DNDC/DayCent parameter sets, emission factors, and the VM0042 / IPCC citations behind them
Email context	Directives, approvals, and context received by email (see D)
Outcomes & timings	Cycle durations and results, to learn what shortens time-to-issuance

C · Write / retrieve / retain

- **Write** on task completion and on every CONFIRM-gated approved action; nothing sensitive is stored outside project scope.
- **Retrieve** the most relevant prior memories (semantic + project-scoped filter) before acting, so proven approaches are reused.
- **Scope & isolate** per project and per agent; cross-project reuse is allowed only for methodology knowledge, never client data.
- **Retain & audit** — every entry timestamped, versioned, and replayable in the audit log; retention follows the project's data-retention policy.
- **Human override** — the Head (John) and the CEO can inspect, correct, or purge any agent memory.

D · EMAIL CONNECTION

The agents' long-term memory is connected to the CEO's mailbox **nitzan@carbonature.io**. Directives, approvals, decisions and context to or from this mailbox are ingested into the relevant agent's memory, so instructions given by email persist and inform future actions. Only project- and department-relevant messages are captured; the CEO can review, correct or purge anything ingested at any time. Additional mailboxes (project inboxes, the MRV Technician, Elad) can be connected under the same rules.

Hardening, scale & cross-cutting

Delivered after Tiers 1 and 2: the non-functional targets, the in-house CRM, and the reference material the developer builds against.

16 Non-functional requirements

All requirements below are **active** and delivered in Tier 3 where the new spec finds them relevant. Several are already met by the built database (noted).

Aspect	Target	Status
Availability	99.5% (business hours) — workflow-driven, not real-time critical	`Tier 3`
Map render	< 2 s for a 500-point project	`met – 83 ms measured`
Excel ingestion	< 30 s end-to-end for < 100 rows	`Tier 1 pipeline`
DNDC run	~30 min for a 150-plot × 6-yr project	`Tier 2 runners`
Storage retention	10+ years; S3 → Glacier after 2 yr	`configured`
Security	Encryption at rest (KMS) & in transit (TLS 1.3); SSO + MFA	`KMS live`
Audit retention	Project lifetime + 5 years; append-only, S3-mirrored	`enforced`
In-house CRM	Built after Tiers 1 & 2; unified with the module's admin; no external vendor	`Tier 3`

A Appendix — VM0042 v2.2 + CCP reference checklist

The compliance engine validates against this checklist; each rule maps to a VM0042 section and is enforced in the built database.

Rule	VM0042 §	Where enforced
Stratified random sampling	§8.2.1.2	Sampling Plan Generator · compliance engine
≥ 5 composite samples per stratum	§8.2.1.2	Plan Generator · `evaluate_compliance()`
3 baseline control sites (QA2)	§8.3	Baseline-definition skill · `bsl_distance_chk` (≤250 km enforced at insert)
9 similarity criteria for BSL (QA2)	Table 7	Project setup · `similarity_criteria` jsonb
30 cm depth, 2 increments	§8.2.1.3	SOC datasheet · `soc_measurements`
ESM-based SOC reporting	§8.2.1.6	dry-mass ÷ probe-area · `esm_soc_stocks`
Same-season sampling window	§8.2.1.1	Plan Generator · Dave's reminders
Dry combustion (Dumas) preferred	§8.2.1.4	Lab integration config
ISO/IEC 17025 or NAPT / GLOSOLAN lab	§8.2.1.4	Admin · `labs` flags
VMD0053 model validation · IME sign-off (QA1)	§8.6.1.3	Model Run Console · `mvr` (IME contracted by VVB)
Monte Carlo uncertainty (QA1)	§8.6.1.2	Uncertainty-Deduction skill · L 500–1000
ICVCM CCP — no DSM for SOC	CCP condition	VT0014 excluded from CCP projects by design

B Appendix — database alignment

This spec is written to a database that is already built and verified — 41 tables in the `mrv` schema, 16 migrations, seven build stages, running on AWS RDS. The screens map onto it as follows.

Module area	Database objects
Spatial core	organizations · projects · farms · plots · strata · sampling_points · baseline_control_sites
Sampling lifecycle	sampling_cycles · work_orders · mcp_tokens · sampling_events · samples
Lab & SOC	labs · lab_imports · soc_measurements · texture_measurements · esm_soc_stocks · import_quarantine
GHG accounting (QA3)	ghg_parameters · fertilizers · machinery_defaults · activity_data · fertilizer_applications · emission_results · leakage · compute_emissions()
Credits & compliance	products · alm_activities · credits · vcu_issuances · compliance_checks · compliance_scores · stratum_statistics · evaluate_compliance()
QA1 models	model_runs · model_results · mvr
Agents & governance	users · project_memberships · agent_memory · agent_action_policies · audit_log · retention_policy
Audit-readiness views	v_sample_chain · v_data_completeness · v_plot_credits · v_real_plots

The GHG engine was verified component-by-component against the calculator workbook (fuel CO_2 + fertiliser N_2O, both farm-years, to four decimals). The compliance engine, the append-only evidence guarantees, and the demo-data interlock are all live. Full detail: the repository's per-stage records and the two-page data-flow / architecture diagrams.